

Behavioural Design in Video Games: Ethical, Legal, and Health Impact on Players

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ABSTRACT

Video games use behavioural design strategies, i.e., dark patterns, to increase engagement and drive revenue. These practices affect consumer behaviour, e.g., lead to extended playtime, and subsequently health, such as social well-being. HCI approaches such as motivational design or personalization are central to behavioural design strategies. Some approaches, e.g., in-game messages to guilt trip users, are ethically and legally questionable. In this workshop, we explore the ethical, health, and legal implications of behavioural design strategies. Our workshop aims to integrate interdisciplinary viewpoints, to co-develop a road map to address behavioural design, and collect contemporary perspectives on the issue. Participants will take away knowledge from different expert perspectives, concrete steps to address the impact of behavioural design, and a multidisciplinary expert network that will tackle existing and emerging future challenges.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

behavioural design, video games, consumer health

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1 INTRODUCTION

The rise of video games spawned a plethora of game genres, experiences, and monetization strategies. Today, game studios face the

challenge of delivering compelling experiences in an overcrowded and competitive market. Success can sometimes yield millions, but failures are frequent and can sink companies. The resulting tension is felt by game creators pursuing their art and craft, and executives mandated to pursue revenue alike. The impact of revenue driven-strategies in turn affects consumers as making good decisions in the context of gaming may become more difficult, e.g., users may stay on a platform longer than intended, reducing their social well-being, or may be pressured into making microtransactions through game mechanics, ultimately affecting their financial health [17].

In the context of the attention economy where engagement is converted into revenue, UX and UI design are key to successful monetization in new technologies, including video games. HCI researchers are at the forefront of engagement with the complexity of interface design, algorithmic design, behaviour change strategies, and motivation-driven design that raises ethical concerns [17]. Design strategies are central to the tension between stakeholders' financial interests, game creators' work, and players' health.

In this workshop, we engage with the positive and negative implications of behavioural design in video games [17]. Starting from 15-minute introductory lectures, we dive into expert working groups discussing 1) the implications of exploiting players' psychology, 2) the tension between publisher needs, game creators' intentions, and consumer needs, and 3) available legal frameworks and policy options, with the goal to derive concrete steps forward to address the negative implications of behavioral design through interface design, education, and policy.

2 BACKGROUND AND MOTIVATION

The European Commission's report "Behavioural study on unfair commercial practices in the digital environment. Dark patterns and manipulative personalization" [13], identifies 22 manipulative design patterns used in commercial games including approaches such as intermediate currency (i.e., the conversion of real money into devalued game money), nagging, forced registration, countdown timer, and hidden information. Practices like "Roach Motel", i.e., easy to get in, hard to get out, or "Immortal Account", i.e., accounts that can not be deleted, demonstrate the aggressiveness of the practices.

Zagal, Björk, and Lewis [20], classified Dark Patterns into temporal (e.g., grinding), monetary (e.g., pay to skip), and social capital-based (e.g., social pyramid scheme) patterns. The scale of how dark

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patterns affect the games market comes into perspective when considering that a) dark patterns are particularly effective when microtransactions are the primary means of monetization, and b) when looking at the role of microtransactions driving revenue growth. Electronic Arts 2020 annual report demonstrates how the company's revenue shifted in 10 years from 6% revenue from live services in 2010 to over 50% in 2020 [1–3]. EA is obviously only a single example, however, when investigating reports from the analyst company Newzoo, a similar trend becomes visible [8]. Further, Zendle et al. [21] illustrate the same trend using open data from Steam, discussing insights into the value of different strategies such as loot boxes, cosmetic microtransactions, and pay-to-win strategies.

The role of HCI research in the development of behavioral design approaches in games is not to be underestimated. HCI work has linked design strategies to motivational theories [7], applied individual differences to optimize engagement [15], and demonstrated the role of customization and identity [4] on engagement and enjoyment. There is a fine line between optimizing user experience through research and applying research insights to malicious in-game monetization practices, requiring product owners and interaction designers to reflect on the ethical implications of their design choices [19].

While the games industry pursues legitimate business, the implications for consumers, including vulnerable populations such as children or people who are less technology literate, are severe enough that policymakers, legal experts, and health organizations have begun to evaluate the implications of regulatory steps toward the industry [13]. The demand for developing broad expertise as interaction designers is exaggerated, because emerging technologies such as AI-driven personalized targeting [5], XR experiences [11], and advanced behavioural design [17] expand the options for malicious behavioral design practices at a speed that is challenging to be regulated by policy, and instead relies on informed decisions by designers keeping the best interest of the user in mind, as well as consumers who need to be empowered to recognize and adequately respond to harmful design practices.

2.1 Workshop Aims

The workshop pursues three goals:

- (1) *Integrating interdisciplinary viewpoints.* Behavioural design is psychologically informed, draws from advanced computational approaches, operates in a societal context defined through legal frameworks, and impacts consumers' well-being in complex ways. The CHI community is a rare opportunity to gather a wide range of researchers to discuss behavioural design.
- (2) *Devising a roadmap to address behavioral design through policy, interaction design, and education.* The topic requires a multi-prong approach that requires insights into what action on a societal level, the user interface level, and the educational level might look like.
- (3) *Creating a state-of-the-art collection of contemporary perspectives.* Perspectives on solutions to behavioural design will vary broadly. We aim to capture these perspectives through submitted content, participant perspectives shared during

the workshop, and a collection of individual interviews conducted by the organizers.

Participants will take away a deeper understanding of the complex ethical and legal implications of behavioural design for players' health, concrete steps to take to address behavioural design in design practice, and a network of individuals interested in addressing the implications of behavioural design in practice.

2.2 Organizers

The organizing team provides an interdisciplinary perspective on the topic, including leading legal experts on children's rights, experts in research on addiction and digital health, and internationally renowned game UX experts. The researchers presented their insights on the implications of behavioural design and the ethics of game design in research contributions [6, 18], independent reports [17], blog posts [9], and keynotes [10].

Max V. Birk is an Assistant Professor in the Department of Industrial Design at Eindhoven University of Technology. Max has investigated the role of video games for mental health, contributed to our understanding of game design elements on experiences and behaviour, and investigated the tension between player expectations, commercial practices, and design strategies.

Simone van der Hof is a Professor of Law and Digital Technologies and Academic Director at the School of Law at Leiden University. She is an internationally renowned expert on children's rights and has led projects focused on the online safety of children, e.g. digital bullying, online privacy, online sexual abuse, and the protection of children from economic exploitation, e.g., children as social media influencers, eSporters, YouTubers, and the online manipulation of children, e.g. new forms of digital marketing, dark patterns.

Celia Hodent has fifteen years of experience in the game industry (Ubisoft, LucasArts, Epic Games, and now as an independent consultant) as a game user experience (UX) professional. She holds a PhD in cognitive psychology and is the author of several books, such as *The Gamer's Brain: How Neuroscience and UX can Impact Video Game Design* & *The Psychology of Video Games*. She is UX Summit adviser at the Games Developer Conference (GDC), and founder and chair of the Game UX Summit. She is a member of the Foresight Committee at CNIL (National Commission on Informatics and Liberty, an independent French administrative regulatory body).

Kathrin Gerling is a Professor of HCI and Accessibility at Karlsruhe Institute of Technology. Her research interests include games and play in sensitive settings, and the impact of design elements on player experience in games and immersive interactive simulations. In the context of this workshop, she is particularly interested in the implications of dark patterns for game accessibility.

Tony van Rooij has over fifteen years of experience in setting up and executing applied and research projects that address these challenges. He gained this experience as a researcher at Ghent University (Belgium), at research agency IVO & Erasmus MC Rotterdam (as a PhD student and researcher), and as a consultant. Within the Trimbos Institute, he focuses on promoting the healthy use of digital technology—digital balance—and video games.

3 WEBSITE

Our call for participation, schedule, organizers, accepted submissions as well as the outputs of the workshop will be presented at <https://behaviouraldesign.my.canva.site/chi-2023>.

4 WORKSHOP

4.1 Pre-Workshop

Collectively we are well-positioned to attract a range of disciplines as workshop participants. The workshop will be advertised through HCI mailing lists, targeted invites, and social media. The interest in ethical technology use is high in and outside of the CHI community. The 2021 workshop “What can CHI do about dark patterns?” [12] successfully attracted a wide audience, we see panels [16], research papers [14], industry key notes [10] and reports [17] being published to outline the complexity of the talk. Considering the current interest to discuss the impact of behavioural design and investigating solutions, the specialist audience we aim to reach, and the thriving games community within SIGCHI, we are confident that the workshop will be well perceived by CHI attendees.

We aim to attract a mixed group of researchers and practitioners working on techniques involved in driving behavioural design, e.g., interface design, AI design, user experience, behavioral change, games research, and experts able to implement change, e.g., designers, policymakers, lawyers, and product owners.

We aim for a workshop size of 30-40 participants, including the organizers. We aim to run three distinct sessions, stimulating 5-7 different perspectives developed by the audience, keeping discussion time and group dynamics in mind.

Before the workshop, participants are invited to submit a short position paper (4-6 pages, ACM Article Template), 1) discussing their perspective on behavioural design in games; 2) relating their expertise to one other discipline, e.g., interaction design and law; and 3) presenting a fictional perspective on advancing behavioral design in today's game design landscape.

4.2 Workshop Structure

Table 1 shows the workshop structure for our 1-day workshop. The workshop is designed as a hybrid event with a focus on co-location with facilitation for 5-7 participants joining remotely. Remote participants will introduce themselves remotely, will be enabled to ask questions after keynote presentations, and will form an online group for discussion and present their insights to the group as one unit.

After welcoming the participants, we will go through a 1-minute per participant presentation session to get a shared sense of participants' backgrounds and topics of interest. Before outlining the structure of the next three blocks, participants will form groups by self-identifying with four different groups: 1) designers, e.g., UX, game design, 2) domain experts, e.g., law, health, 3) technical experts, e.g., algorithmic design, platforms, and 4) policy-enablers, e.g., politicians, policy officers, politicians. Group compositions aim to have a variety of interests within one group; people with multiple heads are asked to take on roles increasing diversity in perspectives during discussions. The workshop is structured in three blocks of 1.5 hours with a 15-minute keynote, 45 minutes for discussion,

Table 1: Workshop Schedule

Start Time	End Time	Program Item
09:00	09:15	Begin of the program
09:15	10:00	Introductions
10:00	10:15	Forming groups
10:15	10:30	Coffee
10:30	10:45	Keynote: Celia Hodent
10:45	11:30	Group work: Ethical Impact
11:30	12:00	Sharing insights
12:00	13:30	Lunch
13:30	13:45	Keynote: Tony van Rooij
13:45	14:30	Group work: Health Impact
14:30	15:00	Sharing insights
15:00	15:30	Coffee #2
15:30	15:45	Keynote Simone van der Hof
15:45	16:30	Group work Legal Impact
16:30	17:00	Sharing insights
17:00	19:00	Break
19:00	21:00	Workshop dinner

and 30 minutes for presentation and discussion. Results will be summarized and made available for the larger HCI community.

Session 1 will begin with a 15-min keynote by Celia Hodent on the ethical implications of exploiting consumer neuroscience in game design. The aim of this session is to discuss the responsibility of game and UX designers in behavioural design practices, and practical ways forward from a designer's perspective.

Session 2 will begin with a 15-min keynote by Tony van Rooij on the tension between publisher needs, designer's intentions, and consumer needs. The aim of this session is to discuss design approaches to serving industrial business claims while keeping consumer needs in mind, with the goal to discuss educational approaches to educate and inform the public, designers, and consumers about the implications of behavioural design.

Session 3 will begin with a keynote by Simone van der Hof focused on current legal frameworks, and national and international initiatives to safeguard children's rights in digital economies. The aim of this session is to stimulate discussion about the relationship between legal frameworks and design practices. The practical outcome of the sessions is a reflection of participants on policy requirements and practical ways forward to protect consumers.

The day will include coffee breaks, lunch, and a workshop dinner aligned with the larger program.

4.3 Post-workshop

The workshop produces insights through authors' submissions, and discussions held during the workshop. The submissions will be made available on the workshop website. We will summarize the submitted articles and discussion for submission to the Interactions magazine. Group discussions will be captured on A1 sheets, summarized as Miro boards, and made available on the workshop website to enable further discussion. In addition, we will schedule in-depth interviews with participants during CHI to record participants' reflections on the behavioral design in games, and stimulate further

discussion. The different streams of insights will be publicized via social media by the different authors and multiplied by the uptake of the workshop participants.

5 CALL FOR PARTICIPANTS

Players' expectations when engaging with games is primarily entertainment through enjoyable in-game experiences. Behavioural design in games leverages psychology and design practice to increase time spent with a product, resulting in tension between consumer interests and business practices, mediated by interaction design.

In this one-day workshop, we aim to bring together designers, domain experts, technical experts, and policy enablers, to discuss the implications and solutions of behavioural design in games on players. We will discuss the implications of behavioural design in three sessions, first focusing on game-design practices, second on the tension between business models, designers' decisions, player expectations, and the context of legal and policy on business choices and consumer experiences.

Applicants are invited to submit a short position paper (4-6 pages) in single-column ACM Master article format. Submissions should include 1) the applicant's perspective on behavioural design in game design; 2) relating their expertise to one other discipline, e.g., interaction design and law; and 3) presenting a fictional perspective on advancing behavioral design in today's game design landscape. We will base acceptance decisions on expertise, paper content, and the range of different perspectives and backgrounds to ensure fruitful discussion.

- Submission deadline: February 23, 2023
- Acceptance notification: March 09, 2023
- Submit to: chi23.behaviouraldesign@gmail.com
- Website: <https://behaviouraldesign.my.canva.site/chi-2023>
- Workshop date & time: TBD
- Location: Hamburg and online

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